SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

Solar Inverter 25kW (PV-INV-SOLAR-25) — Datasheet

1. Overview

Three-phase string PV inverter 25kW, 3 MPPT. Product family: PV inverters. Lifecycle status: **ACTIVE**.

2. Technical specifications

2.1. Ratings

Item	Value
Rated AC output	25 kW
Max DC input voltage	1100 V
MPPT voltage range	200-1000 V
Number of MPPT / strings	3 MPPT / 2 strings each

2.2. Performance and characteristics

Item	Value
Max efficiency	98.6%

2.3. Endurance and communication

Item	Value
Communication	RS-485 Modbus, Wi-Fi dongle option

2.4. Protection and options

Item	Value
Protection degree	IP66
Insulation resistance threshold (default)	100 kΩ

3. Installation data

3.1. Terminals

- Dimensions (W x H x D): 580 x 520 x 230 mm
- Weight: 36 kg
- Degree of protection: IP66
- Operating temperature: -25 to +60 °C (derating above 45 °C)

4. Standards and certifications

Applicable standards:

- IEC 62109-1
- IEC 62109-2
- IEC 61727
- IEC 62116

Certifications held:

- CE
- VDE-AR-N 4105
- KC

Certificates and declarations of conformity are available from the Certification & Compliance team on request. Certifications not listed above are not held for this product; project-specific evaluation can be requested.

5. Ordering information

Item	Value
Ordering code	PV-INV-SOLAR-25
Standard lead time	6 weeks ex-works
Minimum order quantity	1
Warranty	60 months (see POL-WARRANTY)
Firmware	latest 1.90 (see RN-PV-INV-SOLAR-25)

6. Related documents

- IG-PV-INV-SOLAR-25 — Installation and Commissioning Guide
- TG-PV-INV-SOLAR-25 — Troubleshooting and Error Code Guide
- RN-PV-INV-SOLAR-25 — Firmware Release Notes
- FAQ-PV — PV inverters FAQ

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Solar Inverter 25kW (PV-INV-SOLAR-25) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: 500 mm above, 300 mm below and on each side; avoid direct sunlight on the heat sink
- Operating temperature: -25 to +60 °C (derating above 45 °C)
- Degree of protection: IP66
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Measure Voc of each string at the lowest expected temperature
2. Check DC polarity and insulation
3. Select grid code / country setting
4. Verify communication and monitoring
5. Check MPPT operation at noon

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
Insulation fault at sunrise	Firmware defect at humidity > 95%	TG-PV-INV-SOLAR-25 §3.1
Insulation fault after rain	Water ingress in PV connector	TG-PV-INV-SOLAR-25 §3.2
Grid lost alarm	AC breaker tripped / grid outage	TG-PV-INV-SOLAR-25 §3.3
PV over-voltage alarm in winter	String too long for low temperature Voc	TG-PV-INV-SOLAR-25 §3.4

On-site commissioning support can be requested through the Global CS Center. For pv inverters the request is handled by the technical support team.

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Solar Inverter 25kW (PV-INV-SOLAR-25) — Firmware Release Notes

1. Summary

Latest firmware: **1.90**. Updating to the latest version is recommended for all units.

2. Version history

2.1. Version 1.90 (2025-12-08)

- Added: reactive power Q(U) curve
- Fixed: Wi-Fi dongle reconnect failure after router reboot

2.2. Version 1.82 (2025-04-14)

- Fixed: S.01 false insulation fault at sunrise with humidity > 95%

2.3. Version 1.70 (2024-09-09)

- Baseline release

3. Update procedure

1. Back up parameters / project before updating.
2. Connect the engineering tool (v4.6 or later).
3. Do not switch off power during the update.
4. After the update, verify the version and restore parameters if required.
5. If the update fails, do not power-cycle the unit repeatedly; contact support for the recovery procedure.

4. Compatibility

Parameter files from older versions can be loaded; files from newer versions cannot be loaded into older firmware.